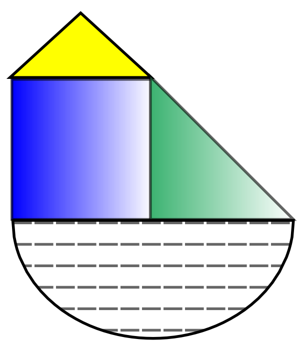
8.2 Area of Composite Figures

A **composite figure** is formed by combining simples shapes. You can calculate the total area of a composite figure by adding or subtracting areas.

My creation:



What Formulas do we Know?

Area of a rectangle: $length \times width$

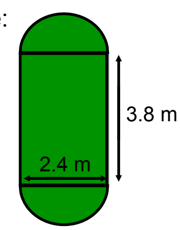
Area of a triangle: $\frac{1}{2}bh$

Area of a circle: πr^2

Perimeter of a circle: $2\pi r$ or πd

Area of Composite Figures

1) Find the area of the following composite figure:

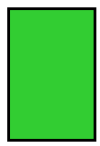


Step 1: Identify the shapes that make up the figure

Step 2: Find the area of each shape

Area of 2 semicircles:

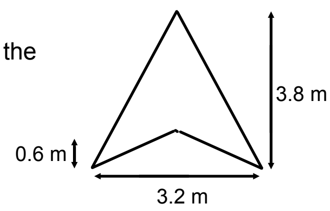
Area of rectangle:



Step 3: Calculate the total area

2) Find the area of the hang glider sail:

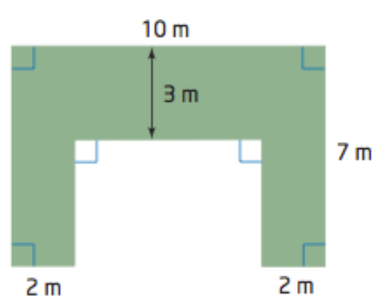
Step 1: Determine how we can find the area of this sail



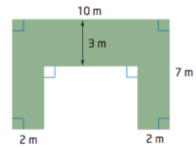
Step 2: Calculate the area of each figure

Step 3: Find the total area

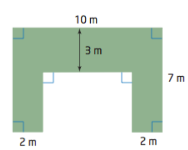
3) **a)** Find the length of the unknown sides for the following object:



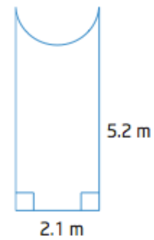
b) Find the perimeter of the object:



c) Find the area of the object:



4) A hotel is remodelling its outdoor entrance area. The new deign includes a tile walkway leading to a semicircular fountain.



a) What steps would you take to find the area of the walkway?

b) Calculate the area of the walkway (round to the nearest tenth)

c) The walkway will have a border in a different colour of tile. Calculate the perimeter of the walkway. (Round to the nearest tenth of a meter)